

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

INFR09027 COMPUTER COMMUNICATIONS AND NETWORKS

Tuesday 18th August 2015

14:30 to 16:30

INSTRUCTIONS TO CANDIDATES

Answer any TWO questions.

All questions carry equal weight.

CALCULATORS MAY BE USED IN THIS EXAMINATION

Year 3 Courses

Convener: S. Viglas

External Examiners: A. Cohn, T. Field

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. (a) List two advantages and two disadvantages of having international standards for network protocols. Name two prominent sets of standards relevant to computer networking. [6 marks]
- (b) The high-quality version of an hour long TV programme on BBC iPlayer has a size of 350MB to download to a home computer. A home user has a broadband service with a download speed of 2Mbps. Round-trip time to the BBC iPlayer website is found to be 100ms using a ping test with default packet size of 56bytes. Assume that the access link is the bottleneck and delays other than those due to propagation and transmission are negligible. Calculate the time it takes for the user to download the whole programme and the throughput of the transfer. How much does the propagation delay matter for this transfer? [6 marks]
- (c) Consider distributing a file of $F = 10$ Gigabits (Gb) to N peers. The server has an upload rate of $u_s = 50$ Mbps, and each peer has a download rate of $d_i = 2$ Mbps and an upload rate of u . For $N = 10$ and 100 , and $u = 250$ Kbps and 500 Kbps, tabulate the minimum distribution times for each of the combination of N and u for both client-server distribution and P2P distribution. [8 marks]
- (d) Describe the Ethernet Media Access Control (MAC) Protocol. [5 marks]

2. (a) Discuss five key differences between circuit and packet switching. [5 marks]
- (b) Suppose that x bits of user data are to be transmitted over a k -hop path in a packet-switched network as a series of packets, each containing p data bits and h header bits, with $x \gg p + h$. The bit rate of the lines is b bps and the propagation delay is negligible. What value of p minimizes the total delay? [4 marks]
- (c) Describe three negative consequences of network congestion. [3 marks]
- (d) Discuss two broad approaches towards congestion control, each with an example. [3 marks]
- (e) Suppose a large organization with 1000 routers employs a simple 2-level routing hierarchy with the routers equally distributed between 20 areas (as in OSPF) to keep the routing table size small. Assume that, unlike OSPF, every router needs to keep an entry in its routing table indicating the route to reach areas other than the one it belongs to. At what network size, in terms of the total number of routers, will it no longer be possible to keep the routing table size at or below 100 entries with the above mentioned 2-level hierarchy? [4 marks]
- (f) How is the notion of “subnets” related to hierarchical routing? [2 marks]
- (g) Give a simple heuristic for finding two paths through a network from a given source to a given destination that can survive the loss of any communication line (assuming two such paths exist and routers never crash). [4 marks]

3. (a) Define multiplexing and demultiplexing at the transport layer. Discuss how demultiplexing differs between TCP and UDP sessions. [3 marks]
- (b) Give five reasons why an application developer might choose UDP over TCP as the underlying transport protocol. [5 marks]
- (c) Outline self-learning in a switch. [1 mark]
- (d) How does self-learning help in the case of a host moving between two access points, both connected to a common switch via two different ports at the switch? [2 marks]
- (e) When a file is transferred between two computers, two acknowledgement strategies are possible. In the first one, the file is chopped up into packets, which are individually acknowledged by the receiver, but the file transfer as a whole is not acknowledged. In the second one, the packets are not acknowledged individually, but the entire file is acknowledged when it arrives. Discuss the relative merits of these two approaches. [4 marks]
- (f) Suppose the data portion (including the headers) of a packet contains a sequence of 8 bytes, each corresponding to the 8-bit unsigned binary representation of the following integers: 243, 81, 128, 64, 32, 16, 27, 9. The corresponding sequence of hex values for those integers is: 0xf3, 0x51, 0x80, 0x40, 0x20, 0x10, 0x1b, 0x09. Compute the Internet checksum for this data. [4 marks]
- (g) Suppose a host has a 1MB file that is to be sent to another host over a 2Mbps link with 10ms one-way propagation delay. The file takes 1 second of CPU time to compress 30%, or 2 seconds to compress 50%. Ignore queueing delays. Calculate the bandwidth at which the combined time taken for compression and transmission for each of the compression options equals that of the uncompressed case. Does propagation delay also get affected? Explain your answer. [6 marks]